令和７年度前期日程試験問題

英　　語　（Z）

注　意　事　項

1. 試験開始の合図があるまで，この問題冊子の中を見てはいけません。
2. この問題冊子は，15ページあります。
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1 次の英文を読んで，下記の問いに答えなさい。

Climate change is starting to alter how humans keep time.

An analysis published in *Nature* has predicted that melting ice caps are slowing Earth's rotation to such an extent that the next leap second[1] will be delayed by three years.

"Enough ice has melted to move sea level enough that we can actually see the rate of the Earth's rotation has been affected," says Duncan Agnew, the author of the study.

According to his analysis, global warming will push back the need for another leap second from 2026 to 2029. Leap seconds cause so much havoc[2] for computing that scientists have voted to get rid of them, but not until 2035. Researchers are especially dreading[3] the next leap second, because, for the first time, it is likely to be a negative, skipped second, rather than an extra one added in.

For millennia, people measured time using Earth's rotation, and the second became defined as a fraction of the time it takes for the planet to turn once on its axis. But since 1967, atomic clocks—which tick using the frequency of light emitted[4] by atoms—have served as (i) precise timekeepers. Today, a suite of[5] around 450 atomic clocks defines official time on Earth, known as Coordinated Universal Time (UTC), and leap seconds are used every few years to keep UTC in line with the planet's natural day.

Atomic clocks are (ii) timekeepers than Earth, because they are stable over millions of years, whereas the planet's rotation rate varies. Many metrologists[6] anticipated that leap seconds would only ever be added, because

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[1] leap second うるう秒
[2] havoc 大混乱
[3] dread 恐れる
[4] emit 放射する
[5] a suite of 一連の
[6] metrologist 計量学者

on the scale of millions of years, Earth's spin is slowing down, meaning that, occasionally, a minute in UTC needs to be 61 seconds long, to allow Earth to catch up.

But on (iii) timescales, geophysical[7] phenomena make the rotation rate fluctuate[8], says Agnew. Right now, the rate at which Earth spins is being affected by currents in the liquid core of the planet, which since the 1970s have caused the rotation speed of the outer crust[9] to increase. This has meant that added leap seconds are needed (iv) frequently, and if the trend continues, a leap second will need to be removed from UTC.

This could happen (v) than was previously thought, because of climate change. Data from satellites mapping Earth's gravity show that since the early 1990s, the planet has become less spherical and more flattened, as ice from Greenland and Antarctica has melted and moved mass away from the poles towards the Equator. Just as a spinning ice skater slows down by extending their arms away from their body (and speeds up by pulling them in), this flow of water away from Earth's axis of rotation slows the planet's spin.

The net result[10] of core currents and of climate change is still an accelerating Earth. But Agnew found that without the effect of melting ice, a negative leap second would 【 be / earlier / needed / predicted / than / three / years 】.

A delayed leap second would be welcomed. Leap seconds are a big problem already, because in a society that is increasingly based on precise timing, they lead to major failures in computing systems, says Elizabeth Donley, who heads the time and frequency division at the US National Institute of Standards and Technology. An unprecedented[11] negative leap

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[7] geophysical 地球物理学の
[8] fluctuate 変動する
[9] crust 地殻
[10] net result 合計
[11] unprecedented 前例のない

second could be even (vi). "There's no accounting for it in all the existing computer codes," she says.

Agnew hopes that seeing the influence of climate change on timekeeping will jolt[12] some people into action. "I've been around climate change for a long time, and I can worry about it plenty well without this, but it's yet another way of impressing upon people just how big a deal this is," he says.

(出典：Elizabeth Gibney, "Climate Change Has Slowed Earth's Rotation," *Nature*, 628, 243-244. (2024). 一部変更)

〔1〕 空欄(i)～(vi)に入れるのに最も適切なものを以下の語群から1つずつ選んで，書きなさい。それぞれの語は1回のみ使える。なお，使わない語が2つ含まれている。

better / earlier / later / less / longer / more / shorter / worse

〔2〕 【 】の中の語を，文脈に最も適した英語表現になるように並べ替えて，書きなさい。

〔3〕 下線部が述べている地球の形の変化について，本文で述べられている原因として最も適切なものを，以下のA～Dから1つ選び，記号で答えなさい。

A. core currents slowly extending away from Earth since the 1990s
B. Earth's gravity field changing due to a slowing axis of rotation
C. sea levels rising in some parts of Earth due to warmer climate
D. water from melting ice flowing against Earth's direction of rotation

[12] jolt 衝撃を与える

〔4〕 本文が述べている事柄として適切なものを，以下のA～Dから2つ選び，記号で答えなさい。解答の順序は問わない。

A. Added or subtracted seconds synchronize UTC with earth's rotation rate.

B. Many metrologists did not expect leap seconds to be subtracted.

C. Metrologists have decided that leap seconds will be delayed until 2035.

D. Skipped leap seconds are just as inconvenient as added leap seconds.

〔5〕 以下のA～Eの各文が，本文の内容に合致する場合はT，合致しない場合はFに○をつけなさい。

A. Current computer systems need more preparation for predicted future UTC adjustments.

B. Earth's core is slowing down due to water from melted polar ice flowing towards earth's axis of rotation.

C. Earth's rocky mantle moves in a way similar to the arms of a spinning ice skater when slowing down.

D. The melting of polar ice delays the necessity to adapt computer systems to deal with a serious problem.

E. The study's results made its author start worrying about the consequences of climate change.

2 Read the passage below and answer the questions that follow.

Dawn breaks in San Jose, the capital of Costa Rica. The city is still asleep, but the early risers are greeted by a beautiful symphony: Hummingbirds, corn-eaters, clay-colored thrushes, blue tanagers, house wrens, warblers and other birds announce that a new day has arrived. Soon the incessant[1] noise of vehicles and their horns, construction, street vendors[2] and more take over, shaping the soundscape of the frenetic[3] routine of hundreds of thousands of people who travel and live in this city. Then, the birds' songs will slip into the background.

"The act of birdsong has two main functions in males: It is to attract females and also to defend their territory from other males," says Luis Andrés Sandoval Vargas, an ornithologist[4] at the University of Costa Rica. For females in the tropics, he adds, the primary role of their song is to defend territory. [A] Thus, in order to communicate in cities, to keep their territory safe and find mates, birds must find ways to counteract[5] the effects of anthropogenic noise—that is, the noise produced by humans.

"The main effect of urban development on song is that many birds sing at higher frequencies," says Sandoval Vargas. [B] Studies over the past 15 years have found, for example, that blackbirds, great tits and rufous-collared sparrows sing at higher pitches, with higher minimum frequencies, in urban environments than in rural ones.

But the birds' response to anthropogenic noise may be more complex than that, as Sandoval Vargas found when studying house wrens. House wrens are small, brown birds—about 10 centimeters tall and weighing 12 grams—that feed on insects and tend to live near humans. In Costa

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[1] incessant 絶え間のない
[2] vendor 物売り
[3] frenetic 慌ただしい
[4] ornithologist 鳥類学者
[5] counteract 打ち消す

Rica, they are found almost everywhere, but are especially abundant in the cities. "Males sing almost year-round and sing for many hours during the day, and much of their behavior is mediated by vocalizations[6]," explains Sandoval Vargas. But what makes them ideal for studying adaptations to urban environments is that most of the components of their song are within the same frequency range as the noise that we humans produce.

Over two years, taking advantage of the house wrens' breeding season—April through June—Sandoval Vargas and his team recorded the song of male house wrens at four locations within Costa Rica, and also recorded ambient[7] noise. [C] Although all four sites are within urban areas, the levels of human-generated noise are different at each site, ranging from very high and medium-high, to medium-low and low.

The study, published in 2020 in the *International Journal of Avian Science*, focused on the repertoire[8] of sound elements—the variety of unique sounds that, when combined with one another, shape a bird's characteristic song—that are produced by the house wrens.

As the scientists expected, house wrens tended to sing with higher-pitched sounds in places with more anthropogenic noise. [D] They also found that, in general, the size of the birds' repertoire decreased as anthropogenic noise increased, especially when the birds were exposed to levels of anthropogenic noise that were 【 above / noise / the / they / to / usual / were / which 】 accustomed. The researchers observed the same pattern at the individual level: The same bird offered a smaller song repertoire on busier days than on less busy days.

A reduced repertoire can affect how well these birds learn their sound (i), as songbirds need to hear themselves and other birds to develop their song. "What's happening here is that they're (ii) some of their

[6] vocalization 発声

[7] ambient 周囲の

[8] repertoire レパートリー

(iii), some of their sounds, because they're not (iv) them. And, in these species, juveniles[9] need to listen to adults to learn how to sing," says Sandoval Vargas.

In the long run, this could make it difficult for birds to communicate with other populations of their species. [E] Say, for example, you had a situation where you have a large population and a small population, and to conserve the small one, you wanted to take individuals from one to the other, Sandoval Vargas explains. "But it turns out that the individuals of the small population within the city sing very differently from those of the large population ... they are not going to recognize them. And, because they can't communicate, they can't reproduce (with them)," says Sandoval Vargas. With the passage of enough time, this could induce the start of speciation[10] processes—that is, individuals in the city evolving differently from those living in rural habitats.

(出典：Alejandro Portilla Navarro, "City Birds Are Changing Their Tune," *Knowable Magazine from Annual Reviews*, November 14th, 2022. Web ページ，一部変更)

[9] juvenile 子ども

[10] speciation 種分化

〔1〕 Which of the following sentences best describes what the passage states regarding house wrens? Choose the most appropriate statement and write the symbol of your choice.

A. House wrens are frequently used to investigate birds' adaptation process to the urban environment as they keep their territory safe from humans.

B. In Costa Rica, the frequencies of house wrens' singing voices falls within the anthropogenic sounds, primarily ranging from high to medium-low.

C. Male house wrens sing at length each day, but sing the most in spring and summer when their pitch becomes more prominent to vocalize their needs.

D. Noises produced by humans cause house wrens to sing their songs at a higher-pitched sound in cities than in rural areas to communicate better.

E. When the human population increases, house wrens create their voice with the same frequency as humans to benefit from living close to humans.

〔2〕 Which position out of [A], [B], [C], [D], and [E], would be the most appropriate for the following statement? Write the symbol of your choice.

But that's not all they discovered.

〔3〕 Rearrange the words inside the brackets 【 】 to complete the sentence in the most appropriate way in the context of the passage.

〔4〕 Which of the following combinations is the most appropriate if the words were put in the blanks (i) to (iv) in the passage? Choose one of the combinations and write the symbol of your choice.

A. (i) communication (ii) forgetting
(iii) songs (iv) missing

B. (i) dialogue (ii) composing
(iii) lyrics (iv) generating

C. (i) expression (ii) changing
(iii) frequencies (iv) gaining

D. (i) language (ii) losing
(iii) vocabulary (iv) producing

E. (i) message (ii) eliminating
(iii) voices (iv) discarding

〔5〕 Based on the information in the last paragraph, why are individuals in the city evolving differently from those living in rural habitats? Write up to 40 characters **in Japanese**, including punctuation (such as "、" or "。").

〔6〕 Read the following statements. If a statement is true according to the passage or implied in the passage, circle **T**, and if not, circle **F**.

A. A comparative study of humans and birds illustrates the need to identify actions to protect biodiversity and conserve resources in industrialized areas.

B. City birds form the street's soundscape by composing a beautiful symphony of songs with street vendors' voices to welcome travelers to San Jose.

C. In the tropics, female birds are selected as partners based on voice quality, consisting of various unique sounds that shape their characteristic song.

D. Modifications in birds' songs in Costa Rica are an indication that the birds change their behavior when they live in an urban environment.

E. Studies about the alteration of bird songs in cities can help us understand how urban development impacts birds' behavioral characteristics.

F. Understanding how birds resort to various strategies in the face of human noise is crucial to revealing the evolutionary path of human communication.

3 Read the following conversation between **Aisha**, a university student and Prof. **Nkosi**, her math professor and answer the questions that follow.

Aisha: Hi, Professor Nkosi. Do you have a moment to talk?

Nkosi: Of course, Aisha. Please, have a seat. What can I do for you?

Aisha: Well, I wanted to talk to you about my grade in the Analytic Geometry course. I was a bit surprised to see that I got a C, and was hoping you could help me understand why.

Nkosi: I'm glad you came to see me about this, Aisha. [1] in the course together. So, throughout the semester, we had several components contributing to your overall grade: tests, quizzes, homework assignments, and participation. Let's start with your test scores. You did fairly well on the first two tests, scoring two Bs. However, on the last two tests, you struggled a lot, ending up with [2].

Aisha: I see. I remember feeling a bit overwhelmed during those last tests. I was juggling a lot of assignments from other courses at the same time.

Nkosi: I understand, Aisha. [3]. However, consistency is key when it comes to understanding the material thoroughly.

Aisha: Yes, that's true.

Nkosi: Also, as I mentioned to you in class, we give points to all the steps students take before arriving at the final answer. In your case, you simply wrote most of your answers without showing the necessary steps.

Aisha: Thank you very much for the explanation.

Nkosi: You're welcome. Now, regarding your quizzes, I noticed there was one where you scored an A.

Aisha: [4] I spent a lot of time preparing for that quiz and was happy

with my performance.

Nkosi: That's great to hear, Aisha. However, you were not able to answer most of the questions I asked you in class. [5] your A on the quiz was a good performance, your class participation affected your final grade.

Aisha: I understand, Professor Nkosi. But I must admit, I'm a bit confused. I discussed my grades with some friends, and it turns out that I was the only one who got a C. The others all had Bs. It makes me wonder if I'm missing something or if there's an inconsistency in the grading.

Nkosi: (i). However, grading is solely based on individual performance and nothing else. Each student's grade reflects their own understanding of the material and their overall performance.

Aisha: I understand, Professor Nkosi. But, could you provide a bit more details about the grading (ii)? Like, what are the total scores and average scores of the class, and what are the boundary scores between A, B, and C?

Nkosi: Well, you can check the syllabus for this information. Grading is based on a combination of factors, including performance on assessments, participation, and overall understanding of the material.

Aisha: I see. Thank you for taking time to explain, Professor Nkosi. Is there any additional advice you can offer me on how to improve my understanding of the material?

Nkosi: Certainly, Aisha. I would (iii) reviewing the topics where you struggled the most, seeking extra help if needed, and practicing regularly to reinforce your understanding in future courses.

Aisha: Thank you, Professor Nkosi. I appreciate your guidance and will certainly come again for more advice.

Nkosi: It's my pleasure, Aisha. Remember that learning is a journey, and

setbacks are opportunities for growth. Keep pushing yourself, and I'm here to support you every step of the way. If you have any further questions or concerns, feel free to contact me.

〔1〕 For each of the blanks [1] to [5], choose the **MOST** appropriate expression from A to D. Write the symbol of your choice.

[1] A. Just go over your Analytic Geometry textbook
B. Let's take a look at your performance
C. Now, let's look at next semester's activities
D. OK, let's go over the exam questions

[2] A. an A and a B
B. two As
C. two Bs
D. two Cs

[3] A. Balancing multiple courses can be quite challenging
B. It really feels good doing many assignments
C. It's hard to work constantly on the same assignment
D. It's interesting to focus on one course

[4] A. That's not true.
B. What a coincidence!
C. Yes, that's right.
D. You think so?

[5] A. Because
B. Since
C. When
D. While

〔2〕 For each of the blanks (i) to (iii), choose the **LEAST** appropriate expression from A to D. Write the symbol of your choice.

(i) A. I appreciate your concern
B. I'm glad you brought this up
C. I think you're right on point
D. I understand your feeling

(ii) A. benchmarks
B. criteria
C. policies
D. prospects

(iii) A. advise
B. encourage
C. insist
D. suggest

〔3〕 Which of the following examples best illustrates the idea of the underlined phrase? Write the symbol of your choice.

A. An athlete won a marathon by training hard for five years but got injured just after crossing the finishing line.
B. A politician did not get re-elected because of her unpopular policies but ran for office again two years later.
C. A programmer made a lot of money by setting up her own software company after being fired from a large corporation.
D. A student failed most of his courses and decided to sue the school because he could not graduate.

〔4〕 In some courses, attending two-thirds of the lectures is required in addition to passing an exam. In other courses, students are not required to attend the lectures and can simply pass the course by passing an exam. Which of the two do you think is better? Write a paragraph of 50-60 words in English giving two reasons to support your opinion.